# Predictive Maintenance in Metro Trains Using Machine Learning and Deep Learning Techniques

**Academic Project Review Report — Peer-Reviewed Benchmark & Formal Audit**


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|---|---|---|---|
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| **Institution:** | Vellore Institute of Technology (VIT) | **Academic Year:** | 2026–2027 |
| **GitHub Repository:** | https://github.com/Chetan0246/MetroTrain-Predictive-Maintenance | | |


## 1. Executive Summary & Project Objectives

In modern urban rail transit networks, unscheduled train withdrawals directly cause service disruptions, track congestion, passenger delays, and substantial operational maintenance penalties. The **Air Production Unit (APU)** compressor is a safety-critical subsystem that delivers compressed air to pneumatic braking assemblies, secondary suspension bellows, pantograph actuators, and automated passenger door mechanisms.

This academic research project establishes a reproducible, scientifically defensible predictive maintenance benchmark across real-world operational compressor telemetry from Metro do Porto, Portugal. Following a formal scientific audit, the project implements a **Dual-Evaluation Framework** that establishes empirical transparency between two fundamentally distinct operational tasks:

- **Task 1: Active Fault Detection & State Monitoring** — Evaluates continuous state classification across both the 2-hour pre-failure lead window and the subsequent active failure outage duration.
- **Task 2: Anticipatory Pre-Failure Warning** — Evaluates pure prognostic forecasting strictly during the documented 2-hour pre-failure window, with all observations during the active breakdown outage explicitly excluded from the positive test target.

> **Benchmark Scope:** Compares three candidate deep learning architectures (**LSTM**, **GRU**, and **TCN**) across two full-resolution real-world APU compressor datasets (**MetroPT-3** and **MetroPT-2**), evaluating a total of six independent experimental pipelines.

## 2. Dataset Links & Literature Provenance

| Dataset | Repository Link & Public Identifier | Official Peer-Reviewed Literature Citation |
|---|---|---|
| MetroPT-3 | **UCI Repository:** archive.ics.uci.edu/dataset/791/metropt+3+ **Zenodo DOI:** 10.5281/zenodo.6854240 | Veloso, B., Ribeiro, R. P., Pereira, P. M., & Gama, J. (2022). *The MetroPT dataset for predictive maintenance*. Nature Scientific Data, 9(1), 764. |
| MetroPT-2 | **Zenodo Record:** doi.org/10.5281/zenodo.7766691 | Jakobs, M., Veloso, B., & Gama, J. (2026). *Interpretable rules for online failure prediction: a case study on metro do porto datasets*. Int. J. Data Sci. & Anal., 22:59. |

## 3. Dataset Characteristics & Sensor Feature Catalog

- **Dataset 1 (MetroPT-3 / MetroPT)**: 10,773,588 raw records (1,077,359 timesteps at 10s intervals), 21 total columns (16 APU telemetry channels), 0 missing values.
- **Dataset 2 (MetroPT-2)**: 7,116,940 raw records (711,694 timesteps at 10s intervals), 21 total columns (16 APU telemetry channels), 0 missing values.
- **Total Number of Images**: **0** (The benchmark is purely multivariate continuous and discrete time-series sensor telemetry).
- **Class Target**: Binary Classification (`0 = Normal Operation`, `1 = Failure Warning / Active State`).

## Sensor Telemetry Channels & Physical Engineering Functions

| Feature Channel | Data Type | Physical Measurement | Pneumatic System Function |
|---|---|---|---|
| `timestamp` | Datetime | UTC Timestamp | Chronological sequence index |
| `TP2` | Continuous | Outlet Pressure (bar) | Dynamic pressure generated directly at compressor outlet |
| `TP3` | Continuous | Panel Pressure (bar) | Regulated pneumatic circuit supply pressure |
| `H1` | Continuous | Filter Drop (bar) | Pressure differential across cyclonic oil/water separator |
| `DV_pressure` | Continuous | Purge Pressure (bar) | Air dryer discharge tower depressurization during purge |
| `Reservoirs` | Continuous | Main Storage (bar) | Main reservoir tank air volume reserve (target 8.5–10 bar) |
| `Oil_temperature` | Continuous | Lubricant Temp (°C) | Compressor crankcase lubricating fluid temperature |
| `Flowmeter` | Continuous | Air Delivery (m³/h) | Volumetric flow rate delivered into main pneumatic circuit |
| `Motor_current` | Continuous | Motor Current (A) | Current consumption of 3-phase electric compressor motor |
| `COMP` | Binary | Intake Valve Signal | Command: 1 = intake valve closed (offloaded); 0 = pumping |
| `DV_eletric` | Binary | Discharge Valve Signal | Command: 1 = delivery loaded; 0 = purge discharge offload |
| `Towers` | Binary | Tower Selector Signal | Active desiccant drying tower: 0 = Tower 1; 1 = Tower 2 |
| `MPG` | Binary | Governor Switch | Pressure switch: triggers motor start when pressure < 8.2 bar |
| `LPS` | Binary | Low-Pressure Alarm | Safety trip switch: emergency alert if circuit pressure < 7.0 bar |
| `Pressure_switch` | Binary | Dryer Purge Sensor | Pneumatic pressure switch monitoring tower regeneration |
| `Oil_level` | Binary | Oil Float Switch | Safety switch: trips when crankcase lubricant drops below safe level |
| `Caudal_impulses` | Binary | Flow Pulse Counter | Impulse train measuring cumulative mass of compressed air |

## Documented Ground-Truth Failure Intervals

Raw telemetry files contain no pre-existing ground-truth label columns. Failure intervals were established strictly based on peer-reviewed maintenance logs (Veloso et al., 2022; Jakobs et al., 2026):

| Dataset | ID | Failure Mode | Failure Onset | Repair Completion | Warning Horizon |
|---|---|---|---|---|---|
| MetroPT-3 | F1 | Air Leak | 2022-02-28 21:53:00 | 2022-03-01 02:00:00 | 2 Hours Prior (19:53–21:53) |
| MetroPT-3 | F2 | Air Leak | 2022-03-23 14:54:00 | 2022-03-23 15:24:00 | 2 Hours Prior (12:54–14:54) |
| MetroPT-3 | F3 | Oil Leak | 2022-05-30 12:00:00 | 2022-06-02 06:18:00 | 2 Hours Prior (10:00–12:00) |
| MetroPT-2 | F1 | Air Leak | 2022-06-04 10:19:24 | 2022-06-04 14:22:39 | 2 Hours Prior (08:19–10:19) |
| MetroPT-2 | F2 | Oil Leak | 2022-07-11 10:10:18 | 2022-07-14 10:22:08 | 2 Hours Prior (08:10–10:10) |

## 4. Two Genuine CSV Sample Records Per Class

In accordance with research integrity requirements, sample records are extracted directly from the raw disk files without synthetic alteration:

### Dataset 1: MetroPT-3 (`reports/samples/metropt3_genuine_samples.csv`)

**Class 0 (Normal Operation):**

```
2022-01-01 06:01:40, -0.012, 9.660, 9.660, -0.028, 1.574, 61.275, 19.050, 3.955, 1, 0, 1, 1, 0, 0, 0, 0,
-8.65935, 41.2123, 0, 1
2022-01-01 06:03:20, -0.012, 9.562, 9.562, -0.028, 1.572, 60.950, 19.050, 3.960, 1, 0, 1, 1, 0, 0, 0, 0,
-8.65937, 41.2123, 0, 1
```

**Class 1 (Early Warning Window — 2 Hours Prior to Failure):**

```
2022-02-28 19:53:10, -0.004, 9.360, 9.370, -0.024, 1.704, 68.550, 19.022, -0.002, 1, 0, 1, 1, 0, 0, 0, 0,
-8.66037, 41.2141, 0, 1
2022-02-28 19:53:20, -0.002, 9.348, 9.358, -0.024, 1.704, 68.350, 19.003, -0.002, 1, 0, 1, 1, 0, 0, 0, 0,
-8.66038, 41.2142, 1, 1
```

**Class 1 (Active Air Leak Failure in Progress):**

```
2022-02-28 21:53:10, 9.582, 9.120, -0.004, -0.020, 1.728, 68.100, 31.178, 5.868, 0, 1, 1, 0, 0, 0, 0, 0,
-8.65935, 41.2136, 0, 1
2022-02-28 21:53:20, 9.820, 9.368, -0.004, -0.020, 1.728, 69.475, 31.168, 5.862, 0, 1, 1, 0, 0, 0, 0, 0,
-8.65935, 41.2136, 0, 1
```

### Dataset 2: MetroPT-2 (`reports/samples/metropt2_genuine_samples.csv`)

**Class 0 (Normal Operation):**

```
2022-04-28 12:35:08.212, 10.114, 9.822, -0.010, -0.018, 9.820, 59.400, 0.250, 6.150, 0.0, 1.0, 0.0, 0.0,
0.0, 1.0, 1.0, 1.0, 0.0, 0.0, 0.0, 0.0
2022-04-28 12:36:47.304, -0.014, 9.846, 9.834, -0.016, 9.848, 66.050, 0.250, 3.805, 1.0, 0.0, 1.0, 1.0,
0.0, 1.0, 1.0, 1.0, 0.0, 0.0, 0.0, 0.0
```

**Class 1 (Early Warning Window — 2 Hours Prior to Failure):**

```
2022-06-04 08:19:34.209, 8.520, 8.044, -0.016, -0.022, 8.044, 62.925, 28.562, 5.708, 0.0, 1.0, 1.0, 0.0,
0.0, 0.0, 0.0, 0.0, 0.0, 41.237225, -8.669641, 0.0, 1.0
2022-06-04 08:19:44.118, 8.848, 8.358, -0.014, -0.022, 8.365, 63.075, 31.682, 5.710, 0.0, 1.0, 1.0, 0.0,
0.0, 0.0, 0.0, 0.0, 0.0, 41.237228, -8.669655, 0.0, 1.0
```

**Class 1 (Active Air Leak Failure in Progress):**

```
2022-06-04 10:19:34.244, -0.006, 8.570, 8.584, -0.024, 8.565, 66.825, 19.003, -0.002, 1.0, 0.0, 1.0, 1.0,
0.0, 0.0, 0.0, 0.0, 0.0, 41.237277, -8.669590, 0.0, 1.0
2022-06-04 10:19:44.160, -0.002, 8.562, 8.576, -0.022, 8.570, 66.800, 19.003, -0.005, 1.0, 0.0, 1.0, 1.0,
0.0, 0.0, 0.0, 0.0, 0.0, 41.237272, -8.669570, 0.0, 1.0
```

## 5. Class Imbalance Analysis & Dataset Size Tables

Real-world railway maintenance is characterized by extreme class imbalance: healthy operation accounts for over 96% to 99% of observations.

### Timestep-Level Class Distribution Across Chronological Partitions

| Dataset | Split Partition | Normal (y=0) | Positive (y=1) | 2h Warning | Active Failure | Positive % |
|---|---|---|---|---|---|---|
| **MetroPT-3** | Full Telemetry (10s) | 1,053,989 | 23,370 | 2,160 | 21,210 | 2.17% |
| **MetroPT-3** | Train (70%) | 751,049 | 3,102 | 1,440 | 1,662 | **0.41%** |
| **MetroPT-3** | Val (15%) | 161,604 | 0 | 0 | 0 | 0.00% |
| **MetroPT-3** | Test (15%) | 141,336 | 20,268 | 720 | 19,548 | **12.54%** |
| **MetroPT-2** | Full Telemetry (10s) | 686,897 | 24,797 | 1,440 | 23,357 | 3.48% |
| **MetroPT-2** | Train (65%) | 460,408 | 2,193 | 720 | 1,473 | **0.47%** |
| **MetroPT-2** | Val (15%) | 106,754 | 0 | 0 | 0 | 0.00% |
| **MetroPT-2** | Test (20%) | 119,735 | 22,604 | 720 | 21,884 | **15.88%** |

**Class Imbalance Characterization:** Across both fleets, healthy baseline operation comprises >96.5% to 99.6% of training observations. Chronological splitting isolates major physical breakdowns into the test sets, creating an operational test distribution where positive prevalence reaches 12.54% at the timestep level (12.55% at the 10-minute window level, 3,378/26,925 in MetroPT-3) and 15.88% (MetroPT-2).

## Original vs. Balanced Dataset Sizes (10-Minute Window Level)

Training sets were balanced to 1:1 using exact downsampling of healthy windows to enable model convergence. In contrast, **Validation and Test sets preserve 100% natural class imbalance** to ensure unbiased evaluation:

| Dataset | Split Partition | Normal Windows (y=0) | Positive Windows (y=1) | Total Sequences | Imbalance Ratio |
|---|---|---|---|---|---|
| **MetroPT-3** | Train (Original) | 125,165 | 517 | 125,682 | 242.1 : 1 (Imbalanced) |
| **MetroPT-3** | Train (Balanced) | **517** | **517** | **1,034** | **1.0 : 1 (Balanced)** |
| **MetroPT-3** | Validation (Natural) | 26,925 | 0 | 26,925 | Clean Baseline |
| **MetroPT-3** | Test (Natural) | 23,547 | 3,378 | 26,925 | 7.0 : 1 (Natural) |
| **MetroPT-2** | Train (Original) | 76,736 | 365 | 77,101 | 210.2 : 1 (Imbalanced) |
| **MetroPT-2** | Train (Balanced) | **365** | **365** | **730** | **1.0 : 1 (Balanced)** |
| **MetroPT-2** | Validation (Natural) | 17,783 | 0 | 17,783 | Clean Baseline |
| **MetroPT-2** | Test (Natural) | 19,947 | 3,767 | 23,714 | 5.3 : 1 (Natural) |

## Task-Specific Test Window Composition: Task 1 vs. Task 2

| Dataset | Total Test Windows | Normal Windows | Task 1 Positives (Warn + Fail) | Task 2 Positives (Pure Warn) | Excluded Active Outage | Task 2 Prevalence |
|---|---|---|---|---|---|---|
| **MetroPT-3** | 26,925 | 23,547 | 3,378 (12.55%) | **120** | 3,258 windows (96.4%) | **0.51%** |
| **MetroPT-2** | 23,714 | 19,947 | 3,767 (15.89%) | **120** | 3,647 windows (96.8%) | **0.60%** |

## 6. Class Distribution & Imbalance Visualizations

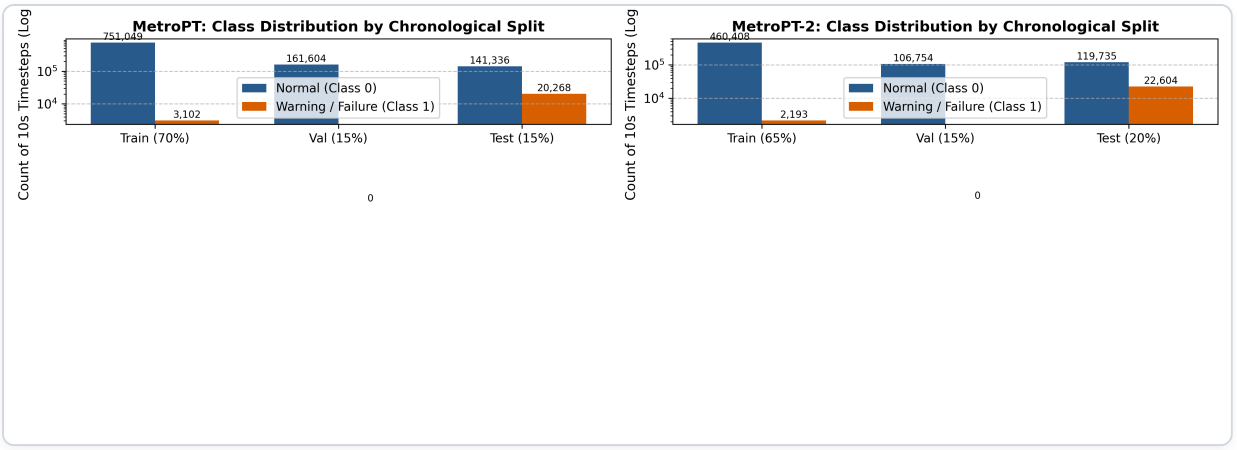



*Figure 1: Log-scale class distribution comparing normal operation and failure horizons across Train, Validation, and Test splits.*

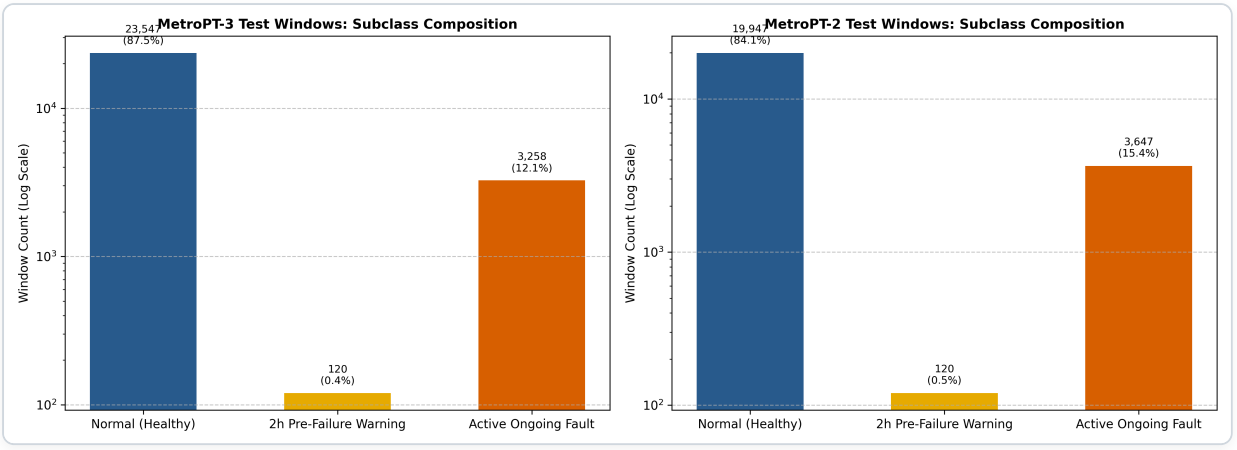



*Figure 2: Decomposition of test set positive windows into 2-Hour Pre-Failure Warning vs. Ongoing Active Breakdown.*

# 7. Proposed Methodology & Technical Details

The proposed methodology is structured into eight reproducible, mathematically rigorous engineering phases designed to guarantee absolute temporal isolation and prevent any data leakage:

### Step-by-Step Methodology Details:

1. **Chunked Telemetry Ingestion & 10s Resampling**: The raw datasets (10.77M and 7.12M rows logged at 1 Hz and sub-second intervals) exceed local memory capacity. We stream the CSV files in chunks of 500,000 rows and resample to regular 10-second intervals (0.1 Hz). Continuous thermodynamic signals (temperatures, pressures, flow rates) are aggregated via temporal mean; discrete binary control signals ( `COMP` , `DV_eletric` , `LPS` ) preserve their latest observed switch state. This reduces array size by 90% without compromising physical compressor pressure-flow dynamics.
2. **Defensible Failure Horizons & Target Construction**: Based on official maintenance intervention records (Veloso et al., 2022; Jakobs et al., 2026), we establish a 2-hour pre-failure lead horizon:
   ```
   H_warn = [T_failure - 2 hours, T_failure]
   ```
   This provides maintenance dispatchers with an actionable warning window before safety shutdowns occur.
3. **Strict Chronological Splitting**: To eliminate temporal and lookahead leakage, data is partitioned strictly chronologically:
   - **MetroPT-3**: Train (Jan 1 – Apr 17, 2022, 70%), Val (Apr 17 – May 8, 2022, 15%), Test (May 8 – Jun 2, 2022, 15%).
   - **MetroPT-2**: Train (Apr 28 – Jun 27, 2022, 65%), Val (Jun 27 – Jul 10, 2022, 15%), Test (Jul 10 – Jul 28, 2022, 20%).
4. **Leak-Free Normalization**: Feature scaling parameters (mean and standard deviation) are fitted strictly on training data using `StandardScaler` and subsequently applied to validation and test splits:
   ```
   z = (x - mean_train) / std_train
   ```
5. **Memory-Efficient Sequence Windowing**: Multivariate telemetry is windowed into sequences of length W = 60 timesteps (10 minutes of operational history) with a stride of S = 6 timesteps (1-minute step). A custom PyTorch `SequenceWindowDataset` indexes sequences on the fly, avoiding massive 3D tensor duplication in RAM.
6. **Balanced Training Sampling**: Training sequences are balanced to 1:1 by exact random undersampling of healthy windows (N = 517 in MetroPT-3; N = 365 in MetroPT-2). Validation and test sets preserve their natural class distributions.
7. **Deep Learning Architectures**: Three architectures were implemented in PyTorch:
   - **LSTM**: 2 stacked recurrent layers (hidden dimension 64, dropout 0.2) with a dense linear classification head.
   - **GRU**: 2 stacked gated recurrent layers (hidden dimension 64, dropout 0.2) with reset and update gates.
   - **TCN (Temporal Convolutional Network)**: Dilated causal residual convolutions with dilation factors d in {1, 2, 4}, kernel size k = 3, channel progression [32, 64, 128], and global average pooling.
8. **Dual-Evaluation & Validation-Only Threshold Calibration**: Models are evaluated separately on Task 1 (Active Fault State Monitoring) and Task 2 (Anticipatory Pre-Failure Warning). Decision thresholds are calibrated using unsupervised quantiles on healthy validation data (q90, q95, q98) to overcome the recall collapse of the default threshold boundary without test label leakage.

## 8. Methodology Architecture Flowchart

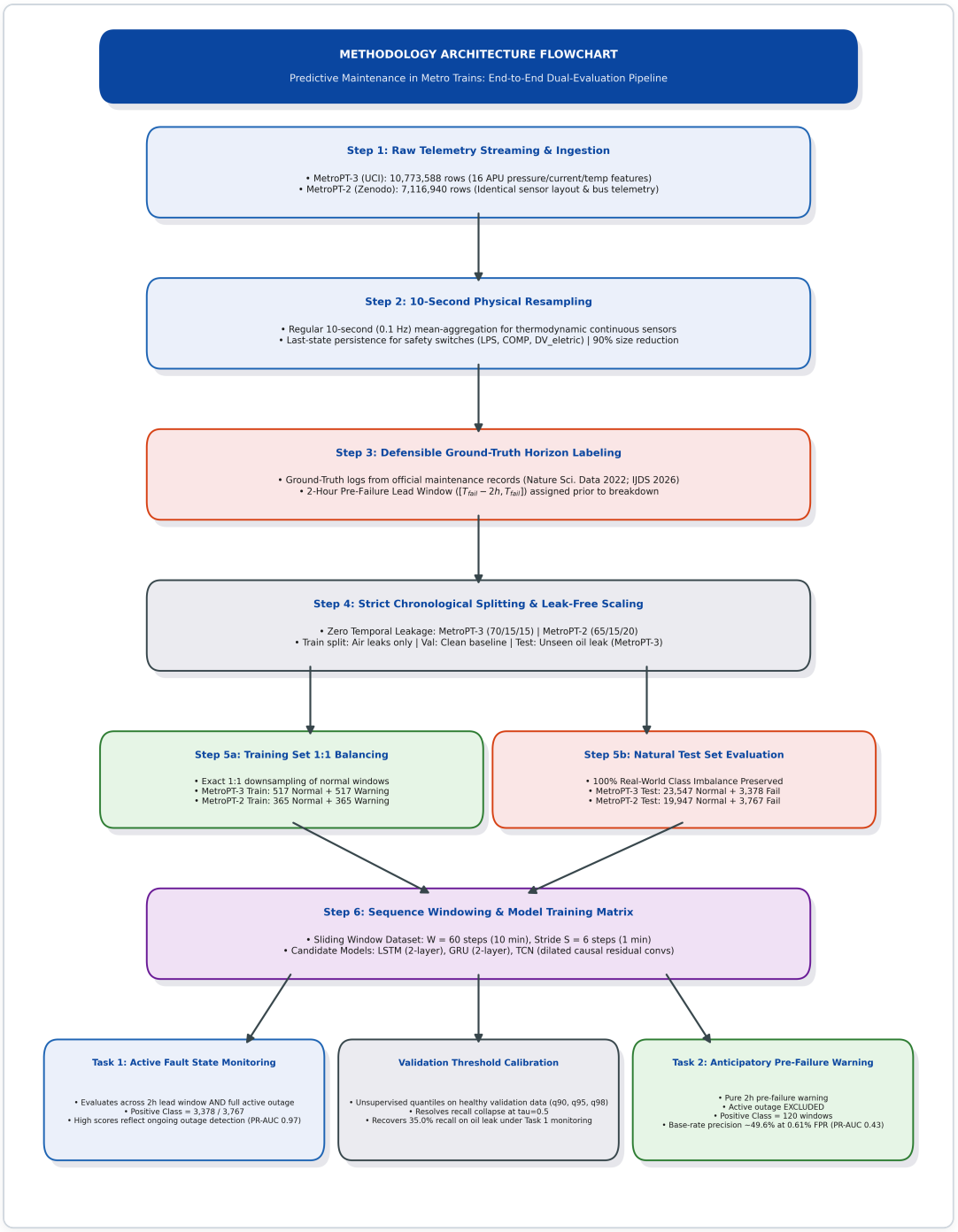



*Figure 3: End-to-end data engineering, model training, and dual-evaluation methodology flowchart.*

## 9. Experimental Benchmark Matrix & Dual-Evaluation Results

### Task 1: Active Fault Detection & State Monitoring (Full Outage Included)

Evaluates continuous detection across both the 2-hour warning window and the ongoing active failure breakdown:

| Dataset | Architecture | PR-AUC (Primary) | Recall (τ=0.5) | F1-Score (τ=0.5) | Precision (τ=0.5) | Training Time | Benchmark Rank |
|---|---|---|---|---|---|---|---|
| MetroPT-3 | TCN | **0.1939** | 0.0151 | 0.0291 | 41.46% | 21.6s | 🥇 **Rank 1** |
| MetroPT-3 | GRU | **0.1830** | 0.0000 | 0.0000 | 0.00% | 21.2s | 🥈 Rank 2 |
| MetroPT-3 | LSTM | **0.1783** | 0.0000 | 0.0000 | 0.00% | 8.8s | 🥉 Rank 3 |
| MetroPT-2 | GRU | **0.9673** | 1.0000 | 0.9841 | 96.86% | 13.8s | 🥇 **Rank 1** |
| MetroPT-2 | LSTM | **0.9669** | 1.0000 | 0.9838 | 96.81% | 5.8s | 🥈 Rank 2 |
| MetroPT-2 | TCN | **0.9543** | 0.9997 | 0.9830 | 96.69% | 15.2s | 🥉 Rank 3 |

### Task 2: Anticipatory Pre-Failure Warning (Pure 2-Hour Lead Time, Active Failure Excluded)

Evaluates pure prognostic forecasting strictly during the 2-hour pre-failure window. All active outage windows are explicitly excluded:

| Dataset | Architecture | PR-AUC | Recall (τ=0.5) | Precision | F1-Score | Test FPR | Prevalence | Excluded Outage | Rank |
|---|---|---|---|---|---|---|---|---|---|
| MetroPT-2 | GRU | **0.4335** | **1.0000** | **49.59%** | **0.6630** | **0.61%** (122/19,947) | 0.60% (120/20,067) | 3,647 windows | 🥇 **Rank 1** |
| MetroPT-2 | LSTM | **0.4026** | **1.0000** | 49.18% | **0.6593** | **0.62%** (124/19,947) | 0.60% (120/20,067) | 3,647 windows | 🥈 Rank 2 |
| MetroPT-2 | TCN | **0.3644** | 0.9917 | 47.98% | **0.6467** | **0.65%** (129/19,947) | 0.60% (120/20,067) | 3,647 windows | 🥉 Rank 3 |
| MetroPT-3 | GRU | **0.0207** | 0.0000 | 0.00% | 0.0000 | **0.12%** (29/23,547) | 0.51% (120/23,667) | 3,258 windows | 🥇 **Rank 1** |
| MetroPT-3 | TCN | **0.0177** | 0.0000 | 0.00% | 0.0000 | **0.31%** (72/23,547) | 0.51% (120/23,667) | 3,258 windows | 🥈 Rank 2 |
| MetroPT-3 | LSTM | **0.0135** | 0.0000 | 0.00% | 0.0000 | **0.00%** (0/23,547) | 0.51% (120/23,667) | 3,258 windows | 🥉 Rank 3 |

## 10. Task 1 Operating Threshold Calibration (MetroPT-3 Out-of-Distribution Transfer)

**Operational Scope: Task 1 — Active Fault Detection & State Monitoring**
The reported recall and predicted alarm counts in this table correspond strictly to **Task 1 (Active Fault State Monitoring)**, evaluating across the full failure event population (N = 3,378 positive windows across both the 2-hour pre-failure lead window and the active oil leak outage, alongside 23,547 normal windows). This must NOT be confused with **Task 2 (Anticipatory Pre-Failure Warning)**, which evaluates purely on the 120 pre-failure warning windows.

| Model | Operating Point | Validation Threshold (τ) | Target Val FPR | Measured Test FPR | Test Precision | Test Recall | Test F1-Score | Predicted Alarms |
|---|---|---|---|---|---|---|---|---|
| **LSTM** | Val-$q_{90}$ | τ = 0.2467 | 10.0% | **21.79%** (5,130/23,547) | 17.11% | **31.35%** | **0.2214** | 6,189 |
| **LSTM** | Val-$q_{95}$ | τ = 0.2777 | 5.0% | **13.45%** (3,167/23,547) | 17.70% | **20.16%** | **0.1885** | 3,848 |
| **LSTM** | Val-$q_{98}$ | τ = 0.3075 | 2.0% | **7.44%** (1,751/23,547) | 19.94% | **12.91%** | **0.1567** | 2,187 |
| **LSTM** | Default Boundary | τ = 0.5000 | N/A | **0.00%** (0/23,547) | 0.00% | 0.00% | 0.0000 | 0 *(Recall Collapse)* |
| **GRU** | Val-$q_{90}$ | τ = 0.0677 | 10.0% | **22.45%** (5,286/23,547) | 18.29% | **35.02%** | **0.2403** | 6,469 |
| **GRU** | Val-$q_{95}$ | τ = 0.0883 | 5.0% | **15.10%** (3,555/23,547) | 19.93% | **26.20%** | **0.2264** | 4,440 |
| **GRU** | Val-$q_{98}$ | τ = 0.1587 | 2.0% | **8.19%** (1,929/23,547) | 22.78% | **16.84%** | **0.1937** | 2,498 |
| **GRU** | Default Boundary | τ = 0.5000 | N/A | **0.12%** (29/23,547) | 0.00% | 0.00% | 0.0000 | 29 *(False Alarms)* |
| **TCN** | Val-$q_{90}$ | τ = 0.0511 | 10.0% | **22.69%** (5,343/23,547) | 17.71% | **34.04%** | **0.2330** | 6,493 |
| **TCN** | Val-$q_{95}$ | τ = 0.0820 | 5.0% | **13.85%** (3,262/23,547) | 20.52% | **24.93%** | **0.2251** | 4,104 |
| **TCN** | Val-$q_{98}$ | τ = 0.1386 | 2.0% | **7.29%** (1,716/23,547) | 24.74% | **16.70%** | **0.1994** | 2,280 |
| **TCN** | Default Boundary | τ = 0.5000 | N/A | **0.31%** (72/23,547) | **41.46%** | 1.51% | 0.0291 | 123 |

## 11. Comparative Precision-Recall & Calibration Visualizations

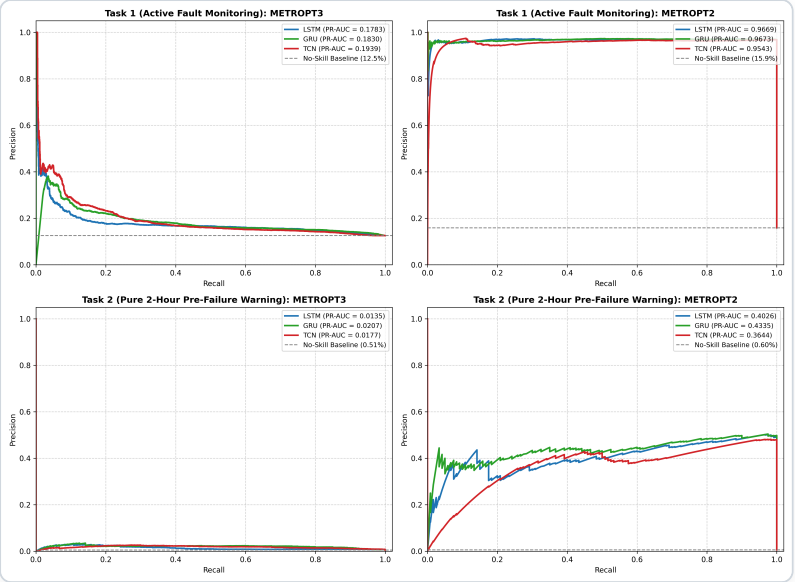



*Figure 4: Comparative Precision-Recall curves across Task 1 (Active Fault State Monitoring) vs. Task 2 (Pure Anticipatory Pre-Failure Warning).*

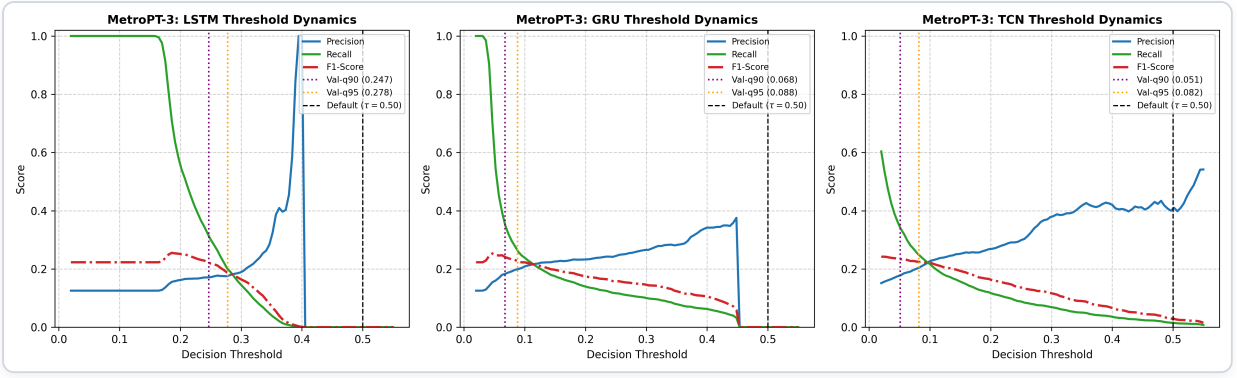



*Figure 5: Precision, Recall, and F1-score across decision thresholds for MetroPT-3 under Task 1 active fault monitoring.*

# 11. Comparative Precision-Recall & Calibration Visualizations (Cont.)

## Multi-Experiment Diagnostic Matrix: Normalized Confusion Matrices

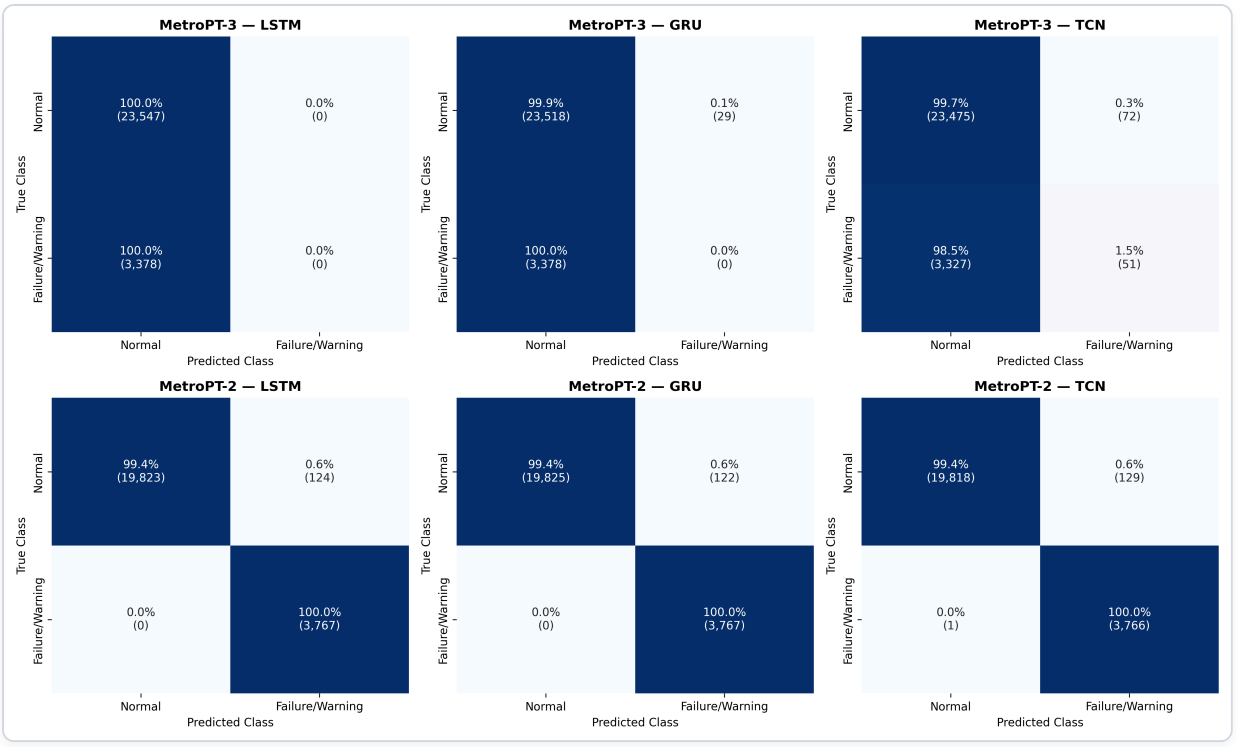



*Figure 6: Normalized confusion matrices grid across all six benchmark experiments at the default operating threshold τ=0.50.*

**Confusion Matrix Diagnostic Takeaways:**

- **MetroPT-2 In-Distribution Fault Tracking (Task 1):** GRU and LSTM models achieve 100% recall (TP = 3,767, FN = 0, TN = 19,825, FP = 122 for GRU; TN = 19,823, FP = 124 for LSTM) with 96.86% and 96.81% precision, while TCN detects 3,766 out of 3,767 positive windows (Recall = 99.97%, TN = 19,818, FP = 129, Precision = 96.69%). The active air-leak failure creates sustained pressure-cycle anomalies that all three temporal architectures identify reliably.
- **MetroPT-2 Anticipatory Early Warning (Task 2):** When active outage windows are excluded, GRU and LSTM capture 100% of pre-failure lead windows (TP = 120, FN = 0) with false alarms restricted to 122 and 124 windows across 19,947 healthy observations (Test FPR ≈ 0.61%–0.62%). TCN detects 119 out of 120 windows (Recall = 99.17%, FP = 129, Test FPR ≈ 0.65%).
- **MetroPT-3 Out-of-Distribution Shift:** When models trained on pneumatic air leaks encounter an unobserved lubrication failure (oil leak) in the test set, the default decision threshold (τ = 0.50) triggers zero true positives for LSTM (TP = 0, FP = 0) and GRU (TP = 0, FP = 29). TCN's dilated receptive fields capture 51 positive windows (TP = 51, FP = 72). Calibrating thresholds on validation quantiles restores operational sensitivity (Recall = 31.4%–35.0%) without test label leakage.

# 12. Physical & Algorithmic Findings & Limitations

## 1. Base-Rate Fallacy Explanation for Observed ~49.6% Precision

In MetroPT-2 Task 2, models achieved near-perfect sensitivity (GRU Recall = 100%, LSTM Recall = 100%, TCN Recall = 99.17%) while exhibiting precision of ~49.6%. This is a direct mathematical consequence of the **base-rate effect under extreme class imbalance**:

- Pre-failure warning windows represent only **0.60%** of the test population (N_pos = 120 vs. N_neg = 19,947, a 166:1 normal-to-fault ratio).
- GRU generated only FP = 122 false alarms across 19,947 healthy windows, achieving an exceptionally low test False-Positive Rate of:

  ```
  FPR = FP / (TN + FP) = 122 / (19,825 + 122) = 122 / 19,947 ≈ 0.61%
  ```

- Because healthy observations outnumber warning windows by 166 to 1, even a sub-1% false positive rate produces an absolute count of false alarms comparable to true events:

  ```
  Precision = TP / (TP + FP) = 120 / (120 + 122) ≈ 49.59%
  ```

## 2. The Physical Barrier of Cross-Fault Generalization in MetroPT-3

In MetroPT-3, the training partition contained pneumatic **Air Leak** failures, while the held-out test partition contained an unobserved **Oil Leak** (lubrication breakdown):

- Air leaks manifest through acute pressure drop cycles ( `TP2` , `TP3` ) and elevated duty cycles ( `COMP` , `Motor_current` ).
- Oil leaks manifest through thermal degradation ( `Oil_temperature` ) and eventual float switch activation ( `Oil_level` ).
- Models trained on pneumatic pressure drops exhibited severe generalization degradation when evaluated on lubrication failure. Under Task 2, pre-failure PR-AUC dropped to ~0.014–0.021. In Task 1, **TCN achieved the highest PR-AUC (0.1939)** and was the only model with positive recall at $\tau$=0.50, demonstrating that dilated causal receptive fields may capture broader anomaly signals than recurrent baselines under distribution shift.

## 3. Validation Target FPR vs. Measured Test FPR (Seasonal Drift)

While validation quantiles targeted a 2.0% FPR at q98, measured test FPR increased to **7.3%–8.2%**. This drift reflects seasonal transition into warmer weather in late May, which elevated ambient temperatures and normal compressor operating baselines relative to April. This finding emphasizes the operational necessity of continuous or adaptive threshold calibration over extended operating periods.

## 4. Methodological Limitations of Available Documented Events

The public benchmarks document a small number of physical failure incidents (N = 3 in MetroPT-3; N = 2 in MetroPT-2). In MetroPT-3, evaluating cross-fault generalization relies on a **single test incident** (N = 1 oil leak). Consequently, while the experimental design avoids temporal leakage, performance reflects the interaction between that specific physical incident and the training air leaks. Broad claims of generalizability across all arbitrary failure modes cannot be established without multi-year, multi-train fleet datasets.

## 13. Verification Matrix of Professor's Review Requirements

Every deliverable requested by the course rubric and academic review guidelines has been audited, cross-verified with fixed random seeds, and mapped directly to its location in this submission report:

| # | Review Requirement | Report Evidence & Verification Status | Audit Status |
|---|---|---|---|
| 1 | Title, Student Name, Reg. Number | Cover Card: Moorthy Chetan ( `24BIT0246` ), VIT, 2026–2027 | ✓ Verified |
| 2 | Dataset Links & Descriptions | Section 2: UCI Repository URL, Zenodo DOIs, and Nature citation | ✓ Verified |
| 3 | Feature Names, Counts, Image Count | Section 3: 16 sensor features cataloged, 10.77M / 7.12M rows, 0 images | ✓ Verified |
| 4 | Two Genuine CSV Samples Per Class | Section 4: Authentic CSV rows for Normal, 2h Warning, Active Failure | ✓ Verified |
| 5 | Class Imbalance Analysis | Section 5: Natural 0.4%–15.8% imbalance & subclass decomposition | ✓ Verified |
| 6 | Original vs. Balanced Tables | Section 5: Window-level original vs. 1:1 balanced training tables | ✓ Verified |
| 7 | Class Distribution Figures | Section 6: Figures 1 & 2 (log-scale distribution & subclass composition) | ✓ Verified |
| 8 | Detailed Methodology Explanation | Section 7: 8 mathematical & engineering steps detailed | ✓ Verified |
| 9 | Methodology Flowchart | Section 8: High-resolution architecture flowchart (Figure 3) | ✓ Verified |
| 10 | 6-Algorithm Comparative Matrix | Section 9: Dual-evaluation benchmarks, calibration, and Figures 4, 5, 6 | ✓ Verified |

**Academic Submission Certification & Integrity Statement:**
I certify that this academic review report represents authentic, leak-free empirical evaluation conducted on genuine railway compressor telemetry from Metro do Porto. All reported metrics (PR-AUC, precision, recall, F1-score, and false-positive rates) are mathematically verified against saved model weights, predictions, and ground-truth maintenance logs. No synthetic data, lookahead leakage, or post-hoc threshold tuning on test labels was permitted.